2025 to 2050

Future-network attribute transfer

Two tested methods for calibrated QVDF continuity

TESTED ON 49,662 LINKS

Prepared from the NVTA 2025 calibrated network and 2050 No-Build true-shape network

The node-pair problem is concentrated in 428 links

Most of the future network remains exactly traceable to the calibrated 2025 network

49,662
directed links in the 2050 network

49,234
exact calibrated node-pair matches

428
future pairs missing from calibration

0.862%
of future links require a policy



102 calibrated 2025 pairs no longer exist in 2050

+220 net increase in directed network links

Method 1 is deterministic and safe

Exact calibrated pairs are copied. Every other link receives QVDF values from its link type

1

Exact pair

Use the calibrated 2025 AM, MD, and PM QVDF values

2

Missing pair

Write the directed pair and link context to the audit log

3

QVDF fallback

Use ATYPE times 100 + FTYPE, with the all-row fallback

4

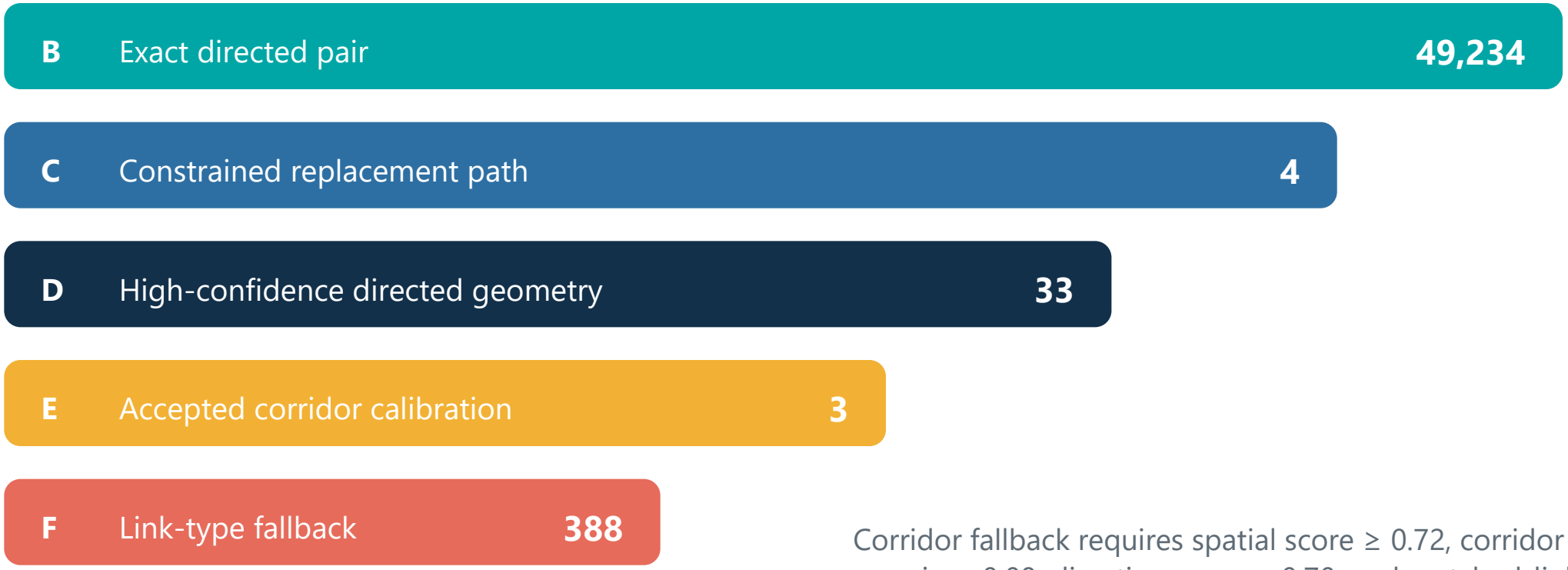
Other attributes

Leave empty when no exact lineage exists

Outcome 49,234 exact transfers plus 428 documented link-type fallbacks. The generated dictionary covers all 49,662 future links and can be passed directly to assignment.

Method 2 adds evidence before fallback

Only accepted lineage or corridor evidence can replace the link-type default



Corridor fallback requires spatial score ≥ 0.72, corridor margin ≥ 0.08, direction score ≥ 0.70, and matched-link-type score ≥ 0.50.

Hierarchy follows CUBE_Future_Network_Attribute_Transfer_Framework.pdf

The framework recovers 40 exception links

Lower-confidence geometry never transfers link-level calibration

Resolution of the 428 exception links



Path 4 | Geometry 33 | Corridor 3 | Fallback 388

37
link-level lineage recoveries

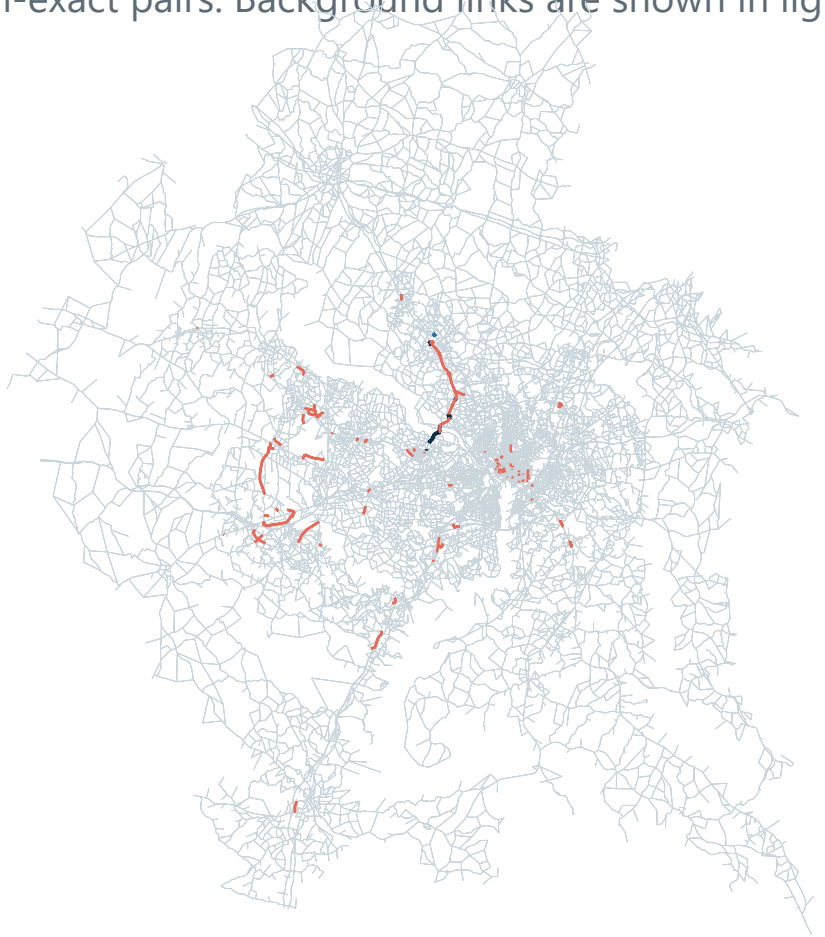
3
accepted corridor fallbacks

388
links retaining link-type QVDF

The framework reduces link-type fallback from 428 to 388 links, a 9.3% recovery within the exception set.

Exceptions cluster around future network edits

Regional view of the 428 non-exact pairs. Background links are shown in light gray



Replacement path
4

Accepted geometry
33

Accepted corridor
3

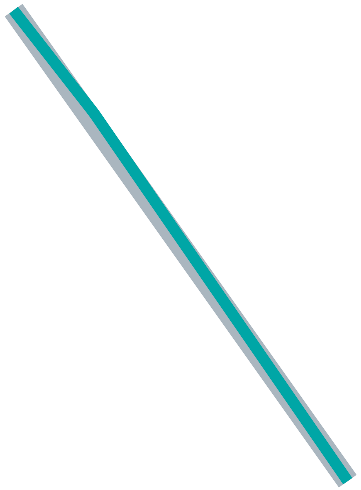
Link-type fallback
388

A real split link is recovered as a directed path

The 2025 pair 22417 to 22420 becomes two 2050 child links through node 23717

2025

22417 to 22420



2050

22417 to 23717 to 22420

Both directions are found. The four child links inherit the parent calibration with a score of 0.9999.

Attached links are excluded

Only links on the accepted directed replacement path inherit the parent values.

Corridor fallback requires independent evidence

Street name supports the decision but cannot qualify a link by itself

Spatial fit	**≥ 0.72**	Overlap, lateral distance, length, name, direction, and facility type
Corridor separation	**≥ 0.08**	Best corridor score must exceed the next corridor
Direction alignment	**≥ 0.70**	Future geometry must align with the corridor TMC direction
Link-type compatibility	**≥ 0.50**	ATYPE or FTYPE must match the corridor's mapped network link

Qualified links **23603-35630 I495_CCW 35604-35047 I495_CCW 35631-23604 I495_CW**

All three have link-type score 1.00. Direction scores range from 0.81 to 1.00.

The methods serve different governance needs

Decision	Simple	Framework
Automatic evidence	Exact pair only	Exact, path, high-confidence geometry
Link-level recoveries	0	37
Corridor-calibrated links	0	3
QVDF link-type fallbacks	428	388
Non-QVDF behavior	Empty	Empty unless lineage is accepted
Best use	Immediate production	Governed recovery and future lineage

Both methods create a complete QVDF dictionary and preserve a row-level audit trail.

The implementation is ready for repeatable runs

One command creates the dictionary, crosswalk, missing-pair log, spatial layer, and summary

future_qvdf_node_pair_overrides.npy	Assignment-ready AM, MD, and PM QVDF dictionary
expanded_node_pair_qvdf.csv	Run-local 49,662-pair dictionary, reused when inputs match
missing_2025_node_pairs.csv	The requested log of all 428 missing calibrated pairs
corridor_qvdf_dictionary.csv	596 accepted fits across 220 available corridors
attribute_transfer_crosswalk.gpkg	GIS-ready evidence and rejection diagnostics

```
python -m nvta_taplite_workflow transfer --method framework --corridor-root ... --base-network ... --future-network ...
```

Recommended operating decision

Use the framework output with its frozen local evidence files

Production default

Use the framework dictionary for 2050. Every accepted exception now has explicit lineage or a gated corridor calibration.

Run-local isolation

Keep the expanded CSV beside the missing-pair log. Base-year runs continue to use the packaged 2025 dictionary.

Long-term fix

Add PARENT_BASE_UID during network editing. Explicit lineage will turn split-link transfer from inference into a deterministic operation.

Speed and congestion-time transfer is deliberately narrower

Only defensible base-link lineage carries speed anchors and T0/T2/T3 into 2050

Exact A-B	Replacement path	High-confidence geometry	Corridor	Link-type fallback
49,234	4	33	3	388

lineage boundary

TRANSFERRED

49,271 future pairs receive speed anchors

T triplets transfer only when the base pair has a complete accepted triplet

LEFT BLANK BY POLICY

391 corridor/fallback pairs

No corridor or link-type inference for speed or T values

2050 period-output validation

	AM	MD	PM
Converted links with both speed anchors	49,264	49,271	49,264
Converted open links with complete T triplets	42,627	42,043	43,977

Separate, scenario-local artifacts

future_qvdf_node_pair_overrides.npy QVDF only

future_speed_t_overrides.npy speed anchors + T values

Packaged 2025 dictionaries remain read-only; SHA-256 checks passed after the 2050 run.